III	Tissue thromboplastin (Tissue factor)	Tissue cells
IV	Calcium ions	Plasma
V	Labile factor (pro-accelerin)	Liver, platelets
VII	Stable factor (pro-convertin)	Liver
VIII	Anti-hemophilic factor	Liver, lung capillaries
IX	Christmas factor Plasma thromboplastin component	Liver
X	Stuart-prower factor	Liver
XI	Plasma thromboplastin anticedent	Liver
XII	Hageman factor	Liver
XIII	Fibrin stabilizing factor	Liver, bone marrow

7.3.1 Causes

Hemophilia is an inherited disease, where clotting occurs at an abnormally slow rate due to the absence of one or more of the blood clotting factors. The sufferers are known as 'hemophiliacs' or 'bleeders'. It is peculiar that it mainly affects males, but in rare circumstances, females can also be affected. There are several types of hemophilia and most forms are inherited. Three major forms of hemophilia namely hemophilia A, B, and C are important.

(i) *Hemophilia A*: Classic hemophilia or Hemophilia A, is a sex linked recessive disorder characterized by a deficiency of factor VIII. About 1 in 10,000 males is born with a deficiency of factor VIII.

(ii) *Hemophilia B*: Hemophilia B is the second most common form of hemophilia. It is due to a dysfunction in factor IX, which is also called as Christmas disease

(iii) *Hemophilia C* : It is mild and is caused by a deficiency of factor XI.

7.3.2 Symptoms

People with hemophilia may lose large amounts of blood from even the smallest injury, and the blood takes a longer time to clot. They will experience spontaneous or internal bleeding and often have painful, swollen joints due to bleeding into the joints. Unusual bleeding after vaccination is also evident. This rare but serious condition can have life-threatening complications. Symptoms of hemophilia vary with the level of clotting factors. If blood clotting-factor level is not very low, bleeding will occur only after surgery or trauma. If blood clotting factor deficiency is severe, it causes spontaneous bleeding.

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